
DP-600: Implementing Analytics Solutions Using Microsoft Fabric

Premium Practice Tests (PDF Resource) | High-Quality Question Bank

Features:

- Total Questions: [117]
- Target Certification: DP-600: Microsoft Fabric Analytic Engineer Practice Exam
- Detailed Explanations: Comprehensive rationale for correct and incorrect options
- Official References: Direct Microsoft Documentation links included for every question
- Branding: Cloud & AI Examprep

Q.1 You have a Fabric tenant that contains a warehouse. The warehouse uses row-level security (RLS). You create a Direct Lake semantic model that uses the Delta tables and RLS of the warehouse. When users interact with a report built from the model, which mode will be used by the DAX queries?

- A) DirectQuery
- B) Dual
- C) Direct Lake
- D) Import

Correct Answer: [A]

Explanation: DirectQuery

Row-level security simplifies the design and coding of security in your application. Row-level security helps you implement restrictions on data row access. Row-level security only applies to queries on a Warehouse or SQL analytics endpoint in Fabric.

When using Direct Lake on SQL endpoints, a query sent to a Direct Lake semantic model can fall back to DirectQuery mode in which case the table no longer operates in Direct Lake mode. It retrieves data directly from the SQL analytics endpoint of the lakehouse or warehouse. Power BI queries on a warehouse in Direct Lake mode will fall back to Direct Query mode to abide by row-level security.

<https://learn.microsoft.com/en-us/fabric/data-warehouse/row-level-security>

<https://learn.microsoft.com/en-us/power-bi/transform-model/desktop-storage-mode#use-the-storage-mode-property>

<https://learn.microsoft.com/en-us/fabric/fundamentals/direct-lake-how-it-works#directquery-fallback>

Why other options are incorrect:

Option B: Dual

The table storage mode can be configured as Import, DirectQuery, or Dual. A table configured as Dual storage mode is both Import and DirectQuery, and this setting allows the Power BI service to determine the most efficient mode to use on a query-by-query basis. Dual mode is typically used in composite models where a table can act as either Import or DirectQuery.

Composite mode can mix Import and DirectQuery modes or integrate multiple DirectQuery data sources. It is not the fallback state for a Direct Lake model encountering RLS.

<https://learn.microsoft.com/en-us/power-bi/connect-data/service-dataset-modes-understand>

<https://learn.microsoft.com/en-us/power-bi/transform-model/desktop-composite-models>

Option C: Direct Lake

Direct Lake is a Power BI semantic model table storage mode option available in Microsoft Fabric. It's optimized for large volumes of data to be quickly loaded into memory from Delta tables available in the OneLake, the single store for all analytics data. Once loaded into memory, the semantic model enables high-performance interactive analysis. This would be the mode if there were no RLS or if the security were manageable by the engine. Since Warehouse RLS is present, Direct Lake is bypassed to maintain security integrity.

<https://learn.microsoft.com/en-us/fabric/fundamentals/direct-lake-overview>

Option D: Import

Import models need to be refreshed, usually on a scheduled basis. A full refresh removes all data from all tables and reloads it from the data source. The entire model must be loaded to memory before Power BI can query the model, which can place pressure on available capacity resources, especially as the number and size of Import models grow.

Model data is only as current as the latest refresh, and so Import models need to be refreshed, usually on a scheduled basis. A full refresh removes all data from all tables and reloads it from the data source. This operation can be expensive in terms of time and resources for the Power BI service and data sources.

<https://learn.microsoft.com/en-us/power-bi/connect-data/service-dataset-modes-understand#import-mode>

Q.2 You have a Fabric tenant that uses a Microsoft Power BI Premium capacity. You need to enable scale-out for a semantic model. What should you do first?

A) At the semantic model level, set large dataset storage format to OFF

B) At the tenant level, set create and use metrics to be enabled

C) At the semantic model level, set large dataset storage format to ON

D) At the tenant level, set Data Activator to enabled

Correct Answer: [C]

Explanation: At the semantic model level, set large dataset storage format to ON

Semantic model scale-out helps Power BI deliver fast performance while your reports and dashboards are consumed by a large audience. Semantic model scale-out uses your Premium capacity to host one or more read-only replicas of your primary semantic model using the large dataset storage format.

By default, scale-out is enabled for your tenant, but it's not enabled for semantic models in your tenant. To enable scale-out for a semantic model, you must use the Power BI REST APIs. Before enabling, the following prerequisites must be met:

1. The Scale-out queries for large semantic models setting for your tenant is enabled (default).
2. Your workspace resides on a Power BI Premium capacity:
 - a. Premium Per User (PPU)
 - b. Power BI Premium P SKUs
 - c. Power BI A SKUs for Power BI Embedded (also known as embed for your customers).
 - d. Fabric F SKUs
3. The Large semantic model storage format setting is enabled.

<https://learn.microsoft.com/en-us/fabric/enterprise/powerbi/service-premium-scale-out>

<https://learn.microsoft.com/en-us/power-bi/enterprise/service-premium-scale-out-configure#enable-scale-out-in-power-bi-service>

Why other options are incorrect:

Option A: At the semantic model level, set large dataset storage format to OFF

This would force the model into the standard legacy format, which does not support the replica architecture needed for scaling out.

<https://learn.microsoft.com/en-us/fabric/enterprise/powerbi/service-premium-large-models>

Option B: At the tenant level, set create and use metrics to be enabled

While usage metrics help you monitor if you need scale-out by checking CPU or Memory usage, enabling them does not technically activate the scaling feature.

<https://learn.microsoft.com/en-us/fabric/admin/tenant-settings-index>

Option D: At the tenant level, set Data Activator to enabled

Data Activator is a no-code event detection engine that transforms data streams into automated actions. It automatically triggers actions when specific patterns or conditions are detected in data sources. It continuously monitors these data sources with low latency, and initiates actions when thresholds are met, or specific patterns are detected. These actions can include sending emails or Teams notifications, launching Power Automate flows, or integrating with third-party systems.

Data Activator is used for real-time alerting and automated actions, e.g. sending an email when a value reaches a threshold. It has no role in memory management or scaling of semantic models.

<https://learn.microsoft.com/en-us/fabric/real-time-intelligence/data-activator/activator-introduction>

Q.3 You are implementing two-dimension tables named Customers and Products in a Fabric warehouse. You need to use slowly changing dimension (SCD) to manage the versioning of data. The solution must meet the requirements shown in the following table.

Table	Change action
Customers	Create a new version of the row
Products	Overwrite the existing value in the latest row

Which type of SCD should you use for each table? To answer, drag the appropriate SCD types to the correct tables. Each SCD type may be used once, more than once, or not at all.

SCD Types	Answer Area
Type 0 Type 1 Type 2 Type 3	Customers: Products:

A) Type 1, Type 3

B) Type 2, Type 1

C) Type 1, Type 0

D) Type 2, Type3

Correct Answer: [B]

Explanation: Type 2, Type 1

Type 2: Slowly changing dimension type 2 is a data warehousing technique that tracks changes to dimension data over time. When a value changes, the system creates a new record with a unique identifier while keeping the original record. This approach preserves a complete history of all changes, so you can see exactly what the data looked like at any point in time.

<https://learn.microsoft.com/en-us/fabric/data-factory/slowly-changing-dimension-type-two>

Type 1: With slowly changing dimension type 1, you either update existing records or add new ones. When a record already exists in your dimension table, you overwrite it with the new values. When it's a completely new record, you insert it as a new row.

This approach keeps only the current state of your data. No historical information is stored. It's perfect for supplementary data like customer email addresses or phone numbers.

<https://learn.microsoft.com/en-us/fabric/data-factory/slowly-changing-dimension-type-one>

Why other options are incorrect:

Option A: Type 1, Type 3

Type 1: With slowly changing dimension type 1, you either update existing records or add new ones. When a record already exists in your dimension table, you overwrite it with the new values. When it's a completely new record, you insert it as a new row.

This approach keeps only the current state of your data. No historical information is stored. It's perfect for supplementary data like customer email addresses or phone numbers.

<https://learn.microsoft.com/en-us/fabric/data-factory/slowly-changing-dimension-type-one>

Type 3: SCD type 3 changes track limited history with attributes. This approach can be useful when there's a need to record the last change, or a number of the latest changes.

This SCD type preserves a limited historical perspective. It might be useful when only the initial and current values should be stored. In this instance, interim changes would not be required.

<https://learn.microsoft.com/en-us/fabric/data-warehouse/dimensional-modeling-dimension-tables#scd-type-3>

Option C: Type 1, Type 0

Type 1: With slowly changing dimension type 1, you either update existing records or add new ones. When a record already exists in your dimension table, you overwrite it with the new values. When it's a completely new record, you insert it as a new row.

This approach keeps only the current state of your data. No historical information is stored. It's perfect for supplementary data like customer email addresses or phone numbers.

<https://learn.microsoft.com/en-us/fabric/data-factory/slowly-changing-dimension-type-one>

Type 0: This type is used for fixed dimensions. Once data is written into a Type 0 dimension row, it can never be changed or overwritten. Any future updates coming from the source data are completely ignored.

<https://radacad.com/scd-slowly-changing-dimension-an-ultimate-guide/>

Option D: Type 2, Type 3

Type 2: Slowly changing dimension type 2 is a data warehousing technique that tracks changes to dimension data over time. When a value changes, the system creates a new record with a unique identifier while keeping the original record. This approach preserves a complete history of all changes, so you can see exactly what the data looked like at any point in time.

<https://learn.microsoft.com/en-us/fabric/data-factory/slowly-changing-dimension-type-two>

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<https://learn.microsoft.com/en-us/fabric/data-warehouse/dimensional-modeling-dimension-tables#scd-type-3>

Q.4 You have a Fabric tenant that contains a lakehouse. You plan to query sales data files by using the SQL endpoint. The files will be in an Amazon Simple Storage Service (Amazon S3) storage bucket. You need to recommend which file format to use and where to create a shortcut. Which **two** actions should you include in the recommendation?

- A) Create a shortcut in the files section
- B) Use the Parquet format
- C) Use the CSV format
- D) Create a shortcut in the tables section

Correct Answer: [B, D]

Explanation: B) Use the Parquet format

You should use the Parquet format for the sales data files because it is optimized for performance with large datasets in analytical processing. Parquet is a columnar file format with optimizations that speed up queries. It's a more efficient file format than CSV or JSON.

<https://learn.microsoft.com/en-us/fabric/data-engineering/lakehouse-overview>

D) Create a shortcut in the tables section

When you create a shortcut in the Tables section of a Lakehouse, Fabric automatically attempts to identify the schema. If those files are in Delta format, Fabric will instantly register them as a managed table. This allows you to query the Amazon S3 data using the SQL Endpoint immediately, without having to move or copy the data.

Shortcuts must be created at the top level (no subdirectories). If the shortcut target contains Delta-formatted data, the lakehouse automatically recognizes it as a table. You can then query it through both Spark and the SQL analytics endpoint.

<https://learn.microsoft.com/en-us/fabric/data-engineering/lakehouse-overview>

Why other options are incorrect:

Option A: Create a shortcut in the files section

Shortcuts let you reference data in a lakehouse without copying it. Instead of ingesting data from another source, you create a pointer that makes the data appear as a local folder or table. Shortcuts can be created at any level of the folder hierarchy. Data in the Files section isn't

automatically registered as a table, but Spark can read it directly for data science or transformation workloads.

If you put your Amazon S3 shortcut in the Files section, the data will remain completely hidden from the SQL analytics endpoint. You would have to manually load them or create a table definition on top of them.

<https://learn.microsoft.com/en-us/fabric/data-engineering/lakehouse-shortcuts>

Option C: Use the CSV format

CSV is a plain text file format that does not support columnar optimization or Delta log structures. The SQL analytics endpoint cannot read or query raw CSV files directly from a shortcut.

<https://learn.microsoft.com/en-us/fabric/data-engineering/lakehouse-overview>

Q. 5 You have a Fabric tenant that contains a lakehouse. You are using a Fabric notebook to save a large DataFrame by using the following code

```
df.write.partitionBy("year","month","day").mode("overwrite").parquet("Files/SalesOrder")
```

For each of the following statements, select Yes if the statement is true. Otherwise, select No

Answer Area

Statement	Yes	No
The results will form a hierarchy of folders for each partition key.	☐	☐
The resulting file partitions can be read in parallel across multiple nodes.	☐	☐
The resulting file partitions will use file compression.	☐	☐

A) No,No,Yes

B) Yes,Yes,Yes

C) No,Yes,No

D) Yes,No,Yes

Correct Answer: [B]

Explanation:

Statement 1: Yes

To optimize throughput, partition your data effectively. Partitioning improves both write throughput and downstream query performance. You can partition data in memory, on disk, or both.

On disk — Use `partitionBy()` to organize data into subdirectories based on column values. Choose columns with good cardinality that produce optimally sized files. Avoid columns that create too many tiny partitions or too few large ones.

In memory — Use `repartition()` or `coalesce()` to distribute data across worker nodes before writing:

`repartition()` increases or decreases partitions with a full shuffle, balancing data evenly.

`coalesce()` only decreases partitions, minimizing data movement.

<https://learn.microsoft.com/en-us/fabric/data-engineering/lakehouse-streaming-data>

Statement 2: Yes

Partitions are the fundamental units of parallelism in Apache Spark and play a crucial role in performance optimization. When you partition 1 GB of data into 100 partitions, Spark attempts to process these 100 partitions concurrently as separate tasks—up to the number of available CPU cores in the cluster (assuming `spark.task.cpus=1`).

<https://learn.microsoft.com/en-us/fabric/data-engineering/spark-best-practices-development-monitoring>

Statement 3: Yes

The effectiveness of compression can affect query performance, as better compression leads to faster data loading and more efficient query processing. Data loading is fast because streaming compressed data boosts transcoding efficiency.

Partitioning data by columns such as year, month, and day, as shown in the DataFrame write operation, organizes the output into a directory hierarchy that reflects the partitioning structure. This organization can improve the performance of read operations, as queries that filter by the partitioned columns can scan only the relevant directories.

Moreover, partitioning facilitates parallelism because each partition can be processed independently across different nodes in a distributed system like Spark. However, the code snippet provided does not explicitly specify that file compression should be used, so we cannot assume that the output will be compressed without additional context.

<https://learn.microsoft.com/en-us/fabric/fundamentals/direct-lake-understand-storage>

Answer Area

Statement	Yes	No
The results will form a hierarchy of folders for each partition key.	☒	☐
The resulting file partitions can be read in parallel across multiple nodes.	☒	☐
The resulting file partitions will use file compression.	☒	☐
